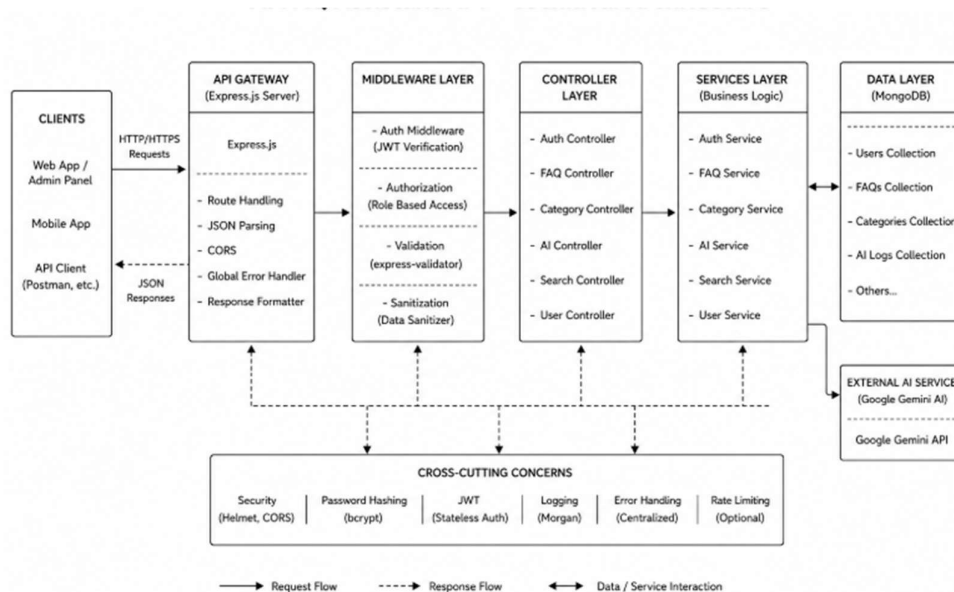


PHASE-2

Hardware Requirements

- **Processor: Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.**
- **RAM: 8 GB minimum (16 GB recommended for concurrent instances of MongoDB processing layers).**
- **Storage: 1 GB of available disk workspace.**

Epic-1: Project Architecture



The AI FAQ Assistant application exposes structured endpoints handled by a layered server architecture model. Incoming requests undergo filter processing before interacting with data services.

Components Definition

- **Interface: API tools like Postman query distinct route paths to extract JSON entities.**

- **Express Server Gateway:** Monitors network ports, maps explicit HTTP methods (GET, POST, PUT, DELETE), parses incoming JSON payloads, and handles CORS headers.
- **Authentication Middleware:** Extracts tokens from headers, validates user context signatures, and evaluates authorization thresholds for guarded resource paths.
- **Granular Services (Logic Modules):** Enforces isolated domain rules for user session life cycles and operational query modifications.
- **Database Interface Layer:** Mongoose translates operational application logic into performant native MongoDB queries.
- **AI Flow (Service Integration):** Connected to external AI models via Google Gemini SDK integrations, letting the controller layer offload complex question-answering tasks and automated topic generation workflows.